

A CANONICAL ORDER-INTERVAL REDUCTION FOR QUESTION 1.2 OF ARXIV:2404.05419

AGENT_LANE_15

SOURCE AND TARGET QUESTION

The source paper is Jose Rodriguez, *Dunford-Pettis type properties in L_1 of a vector measure*, arXiv:2404.05419. In the introduction Rodriguez asks the following question.

Suppose that $L_1(\nu)$ has the positive Schur property. Does $L_1(\nu)$ have the Dunford-Pettis property?

stating that $L_1(\nu)$ has the Dunford-Pettis property if it has the positive Schur property and ν has σ -finite variation (Corollary 4.5). It seems to be an open question whether the assumption on the variation can be dropped, namely:

Question 1.2. *Suppose that $L_1(\nu)$ has the positive Schur property. Does $L_1(\nu)$ have the Dunford-Pettis property?*

Finally, in Example 4.6 we discuss a class of vector measures ν such that $L_1(\nu)$ has the positive Schur property and the Dunford-Pettis property, but fails to be lattice-isomorphic to an AL-space, among other interesting properties.

DEFINITIONS AND SOURCE FACTS USED

Let X be a Banach space, let (Ω, Σ) be a measurable space, and let $\nu \in \text{ca}(\Sigma, X)$. Write $E = L_1(\nu)$ and let

$$j_\nu : L_\infty(\nu) \longrightarrow L_1(\nu)$$

be the canonical inclusion. Rodriguez notes in Section 2 that

$$K_\nu := j_\nu(B_{L_\infty(\nu)}) = [-\chi_\Omega, \chi_\Omega]$$

is a weakly compact order interval in E .

A bounded subset K of a Banach space E is a Dunford-Pettis set if every weakly null sequence (φ_n) in E^* converges uniformly to zero on K , that is,

$$\sup_{x \in K} |\varphi_n(x)| \longrightarrow 0.$$

We use the standard sequential characterization of the Dunford-Pettis property: E has the Dunford-Pettis property if and only if

$$\varphi_n(x_n) \longrightarrow 0$$

for all weakly null sequences (x_n) in E and (φ_n) in E^* . This is the criterion quoted in Rodriguez before Theorem 4.1.

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We also use two preliminary facts recorded by Rodriguez. First, a Banach lattice has the positive Schur property if and only if every relatively weakly compact subset is L-weakly compact. Second, for subsets of $L_1(\nu)$, L-weak compactness is equivalent to approximate order boundedness: for every L-weakly compact $F \subset L_1(\nu)$ and every $\varepsilon > 0$, there is $\rho > 0$ such that

$$F \subset j_\nu(\rho B_{L_\infty(\nu)}) + \varepsilon B_{L_1(\nu)}.$$

These are Propositions 2.4 and 2.5 of the source paper.

MAIN REDUCTION

Theorem 1. *Let $\nu \in \text{ca}(\Sigma, X)$ and suppose that $E = L_1(\nu)$ has the positive Schur property. Then the following are equivalent.*

- (1) *E has the Dunford-Pettis property.*
- (2) *$K_\nu = j_\nu(B_{L_\infty(\nu)}) = [-\chi_\Omega, \chi_\Omega]$ is a Dunford-Pettis subset of E .*
- (3) *The adjoint $j_\nu^* : E^* \rightarrow L_\infty(\nu)^*$ is a Dunford-Pettis operator.*

Proof. First assume that E has the Dunford-Pettis property. Since K_ν is weakly compact, it is a Dunford-Pettis set. For completeness, here is the short sequential argument. Let (φ_n) be weakly null in E^* and suppose uniform convergence on K_ν fails. Passing to a subsequence, choose $u_n \in K_\nu$ and $\delta > 0$ with $|\varphi_n(u_n)| \geq \delta$ for all n . By weak compactness and Eberlein-Smulian, a further subsequence (u_{n_k}) converges weakly to some $u \in E$. Then

$$\varphi_{n_k}(u_{n_k}) = \varphi_{n_k}(u_{n_k} - u) + \varphi_{n_k}(u).$$

The second term tends to zero because (φ_{n_k}) is weakly null, and the first tends to zero by the Dunford-Pettis property applied to the weakly null sequences $(u_{n_k} - u)$ and (φ_{n_k}) . This contradicts $|\varphi_{n_k}(u_{n_k})| \geq \delta$.

Conversely, assume that K_ν is a Dunford-Pettis set. Let (f_n) be weakly null in E and let (φ_n) be weakly null in E^* . Put $M = \sup_n \|\varphi_n\| < \infty$. The set $F = \{f_n : n \in \mathbb{N}\}$ is relatively weakly compact. Since E has the positive Schur property, Rodriguez's Propositions 2.4 and 2.5 imply that for every $\varepsilon > 0$ there is $\rho > 0$ such that

$$F \subset j_\nu(\rho B_{L_\infty(\nu)}) + \varepsilon B_E.$$

For each n , choose $h_n \in j_\nu(\rho B_{L_\infty(\nu)}) = \rho K_\nu$ with $\|f_n - h_n\| \leq \varepsilon$. Since scalar multiples of Dunford-Pettis sets are Dunford-Pettis sets,

$$|\varphi_n(h_n)| \leq \sup_{h \in \rho K_\nu} |\varphi_n(h)| \longrightarrow 0.$$

Therefore

$$\limsup_n |\varphi_n(f_n)| \leq \limsup_n |\varphi_n(h_n)| + M\varepsilon \leq M\varepsilon.$$

As $\varepsilon > 0$ is arbitrary, $\varphi_n(f_n) \rightarrow 0$. The sequential criterion gives the Dunford-Pettis property of E .

Finally, (2) and (3) are the same condition written in dual operator form. For every $\varphi \in E^*$,

$$\|j_\nu^* \varphi\| = \sup_{g \in B_{L_\infty(\nu)}} |\varphi(j_\nu g)| = \sup_{u \in K_\nu} |\varphi(u)|.$$

Thus K_ν is a Dunford-Pettis set exactly when j_ν^* maps weakly null sequences in E^* to norm-null sequences in $L_\infty(\nu)^*$. $\square$

CONDITIONAL CONSEQUENCE FOR THE OPEN QUESTION

Corollary 1. *A positive answer to Rodriguez’s Question 1.2 is equivalent to the following canonical interval assertion: whenever $L_1(\nu)$ has the positive Schur property, $j_\nu(B_{L_\infty(\nu)})$ is a Dunford-Pettis subset of $L_1(\nu)$.*

Proof. If the canonical interval assertion holds, the theorem gives the Dunford-Pettis property. Conversely, if Question 1.2 has a positive answer, then every such $L_1(\nu)$ has the Dunford-Pettis property, and the theorem gives that the canonical interval is a Dunford-Pettis set. $\square$

RELATION TO THE SIGMA-FINITE VARIATION CASE

Rodriguez proves in Corollary 4.5 that the answer is positive when ν has σ -finite variation. In the language of the reduction above, Proposition 4.4 of the source paper verifies the canonical interval condition in that case: sequences in $j_\nu(B_{L_\infty(\nu)})$ are bounded and equi-integrable, so weakly null functionals in $L_1(\nu)^*$ converge to zero on paired sequences from that interval.

LIMITATIONS AND NOVELTY CHECK

This packet is not a full solution of Question 1.2. It proves an exact reduction: after assuming the positive Schur property, the full question is equivalent to proving that one specific weakly compact order interval is a Dunford-Pettis set. The remaining dependency is isolated, but not proved here.

Cheap run-index searches for arXiv:2404.05419, the source title, the phrase “positive Schur property”, and the exact wording of Question 1.2 found no existing solution packet for this target. A bounded web search for the exact question and close variants returned the source paper and adjacent Banach lattice literature, but no separate resolution of Question 1.2.

HUMAN REVIEW RECOMMENDATION

The argument is short and should be reviewable directly. The main points to check are:

- the use of Rodriguez’s approximate order-boundedness characterization;
- the standard implication that weak compact subsets of a Dunford-Pettis space are Dunford-Pettis sets;

- the operator wording: the interval condition is equivalent to j_ν^* being Dunford-Pettis, not to j_ν being Dunford-Pettis.

REFERENCES

- [1] J. Rodriguez, *Dunford-Pettis type properties in L_1 of a vector measure*, arXiv:2404.05419, 2024.
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- [4] P. Meyer-Nieberg, *Banach Lattices*, Springer, 1991.